Elective Bidding System

1. Functional Requirements (FR)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	System shall allow students to distribute 100 bidding credits across ranked elective preferences and validate prerequisite completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core allocation mechanism; prevents invalid enrollment.	Foundation requirement for the bidding process.
FR-002	Functional	System shall allow students to submit, edit, and withdraw bids before the round closes.	High	Pass: Edits/withdrawals update records and credits correctly before deadline. Fail: Bids modifiable after close, or credits not re-allocated on withdrawal.	Gives students flexibility over their bids.	Requires clear deadline enforcement.
FR-003	Functional	System shall enforce max/ min class size and close bidding once seat capacity is reached.	High	Pass: No course exceeds defined capacity. Fail: Course allocated beyond capacity, or under minimum	Prevents overbooking; protects class quality.	Min-size rule may need an admin override path.
FR-004	Functional	System shall notify students of final allocation via email and in-app alert within 24 hours of round closing.	Medium	Pass: All students notified within 24 hrs with correct details. Fail: Notification delayed or contains incorrect data.	Timely communication reduces support burden.	Consider SMS as a backup channel.
FR-005	Functional	System shall provide a waitlist for oversubscribed electives, ranked by bid amount then submission timestamp.	Medium	Pass: Waitlist order correct; freed seats auto-offered. Fail: Order incorrect, or seats not offered to next eligible student.	Reduces manual admin intervention for reallocation.	Define response window for waitlist offers.

2. Non-Functional Requirements (NFR)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
NFR-001	Performance & Security	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures solver scales for institution-wide rounds.	Benchmark with realistic bid distributions.
NFR-002	Scalability	The system shall support at least 2,000 concurrent users submitting or editing bids without response time exceeding 3 seconds per action.	High	Pass: Load testing confirms response times stay under 3 seconds at 2,000 concurrent sessions. Fail: Response times exceed threshold or errors occur under load.	Bidding rounds see spikes near deadlines.	Load test near-deadline conditions specifically.

3. Use Case Flow: Elective Submit Bid

Primary Actor: Student
Preconditions: Bidding round is open; student is authenticated; student has not exceeded the maximum number of active bids.
Postconditions(Success): Bid is recorded with allocated credits and included in the allocation run

Main Success Scenario:
1. Student logs into the bidding portal and selects “Elective Bidding”
2. System displays eligible electives, remaining seats, and the student’s prerequisite completion status for each course.
3. Student ranks preferred electives and allocates a portion of their 100 credits to each ranked choice.
4. Student submits the bid form
5. System validates that total allocated credits do not exceed 100 and that prerequisites are met for each selected elective
6. System deducts the allocated credits, records the bid, and displays a confirmation screen summarising the submission.
7. System sends a confirmation notification (email and in-app) to the student.

Postcondition: Bid is recorded and included in the next allocation run.

Alternate Flow A1 — Credit Allocation Exceeds Limit

Trigger: At step 5, total allocated credits exceed 100.

1. System detects that allocated credits exceed the 100-credit limit.
2. System rejects the submission and displays an inline error: "Total allocated credits exceed your available balance. Please adjust your allocation."
3. System highlights the offending fields and preserves the student's entered values.
4. Student adjusts the credit allocation.
5. Flow resumes at Main Success Scenario step 4.

Postcondition: No bid is recorded until total allocated credits are within the 100-credit limit.

Alternate Flow A2 — Elective Reaches Capacity (Waitlist)

Trigger: At step 5, a selected elective has reached its seat capacity.

1. System detects the elective is at full capacity (extends Submit Elective Bid with Waitlist Placement).
2. System offers the student the option to place the bid on the course waitlist instead.
3. Student confirms waitlist placement.
4. System records the bid as waitlisted, ranked by bid amount and submission timestamp.
5. Flow resumes at Main Success Scenario step 6, with confirmation noting waitlisted status.

4.Use Case Diagram

